The Next Generation Sequencing Quality Initiative

The Next Generation Sequencing (NGS) Quality Initiative is a collaboration between the Centers for Disease Control and Prevention (CDC), the Association of Public Health Laboratories (APHL), and state and local public health laboratories (PHLs) to address the many challenges laboratories encounter when implementing NGS-based assays. The Initiative is developing an NGS-focused quality management system (QMS) to assure foundational quality during the development and implementation of sequencing-based tests by providing customizable, ready-to-implement tools and resources that laboratories can use to standardize and institute quality management practices and procedures. The NGS Quality Initiative has published additional tools and resources, including templates and procedures, that may be of assistance to laboratories throughout their NGS workflow. Please visit the following website to access these resources: https://www.cdc.gov/labquality/qms-tools-and-resources.html.

This document is intended to be used as a tool for implementing, improving, or maintaining an NGS QMS. Blue text provides examples for appropriate input and can be changed, deleted, or augmented as needed for the laboratory's specific requirements.

These documents and tools are not controlled files; format and content **must** be modified as needed to meet the document control, QMS, or regulatory requirements within your laboratory. It is the responsibility of your laboratory to take any necessary actions to ensure the information within these documents remains applicable.

Insert Laboratory-Specific Name Here
NGS Method Validation Summary Report
Template

1.0 Scope

0.1 **Method Validation Report for:** (insert method name)

0.2 **Branch/Laboratory:** (branch and laboratory name)

0.3 **Test System Name:** (insert test system name and indicate whether new or existing test system)

0.4 **Optional: Test Order Code (Client services manual):** (Test code)

0.5 **Test Procedure Document Number(s) and Revision Number:**

Document Name	Document Number	Revision Number	Effective Date
[insert laboratory-specific document name here]	*[insert laboratory-specific document n umber here]*	*[insert laboratory-specific revision number here]*	*[insert laboratory-specific effective date here]*

2.0 Changes from the method validation plan, as applicable:

If new document versions were created, provide details describing the change. Examples include changes to the test method, changes to the number, or origin of samples. If new document versions were created, record them in the table.

Document Name	Document Number	Revision Number	Effective Date
[insert laboratory-specific new version document name here]	*[insert laboratory-specific new version document n umber here]*	*[insert laboratory-specific revision number of the new version here]*	*[insert laboratory-specific effective date of the new version here]*

3.0 Summary of the Test Procedure Purpose/Principle:

1.0

2.0

2.1 **Assay result type** (choose one):

Qualitative

Qualitative-titered

4.0 Specimen Condition Parameters:

3.0

3.1 Information to support stability for submitted isolates: If applicable, the table below is used to reference related documentation for isolate stability.

Test Procedures	Submitter Guidance	Isolate Non-Viability Metrics	Literature
Add rows as necessary, if applicable	(Incubate the inoculated agar slants overnight at 37°C with 5% carbon dioxide to ensure viability of the isolate. Ship at room temperature).	Describe quality indicator (e.g., growth/no growth), refer to SOP.	(References)

5.0 Summary of Results:

4.0

4.1 For minimum acceptable values, refer to acceptance criteria provided i section 6.0 in the Method Validation Plan; if values are not met, but the test process is revised, this should be detailed below in section 6.0.

TP- True Positive, TN- True Negative, FP- False Positive, FN- False Negative, CV-Coefficient of Variation

Characteristic	Number of samples tested	**Actual Performance** (complete calculations)	Acceptable value (if pre-defined)
Accuracy	(# of positive samples and # of negative samples) were measured.		Enter the value, or “n/a”
Precision/ Reproducibility (Qualitative)	(# of positive samples and # of negative samples) were measured in (#) separate runs over at least (# ≥ 3) days, at least (#) days apart, by (#) different operators.		Enter the value, or “n/a”

Characteristic	**Number of samples tested**	**Actual Performance** (complete calculations)	**Acceptable value (if pre-defined)**
Precision/ Reproducibility (Raw Value) (e.g., ANI score, CT value)	(# of positive samples and # of negative samples) were measured in (#) separate runs over at least (# ≥ 3) days, at least (#) days apart, by (#) different operators.		Enter the value, or "n/a"
Clinical Sensitivity/ Diagnostic Sensitivity	(# of positive clinical samples) were measured.		Enter the value, or "n/a"
Analytic Sensitivity (Qualitative)	(# of samples of different serotypes/ genotypes/ geographical variants/species expected to be detected) were measured (# of quantifiable samples at specified concentration levels near the expected detection limit) were measured.		Enter the value, or "n/a"
Clinical Specificity/ Diagnostic Specificity	(# of negative clinical samples) were measured.		Enter the value, or "n/a"
Analytic S pecificity (Qualitative)	(# of negative samples known to contain potentially cross-reactive agents or analytes) were measured.	With description of cross-reactive agents or analytes	Enter the value, or "n/a"

Characteristic	Number of samples tested	Actual Performance (complete calculations)	Acceptable value (if pre-defined)
Limit of Detection	Depth of coverage range of (# to #) and consensus (% to %) were used to determine the bioinformatics LOD. Target biological material present in the range (# to #) were used to determine the biological LOD (e.g., testing/ sequencing of serial dilutions).	Bioinformatics LOD = (minimum depth of coverage and consensus %) Biological LOD = (minimum amount of target biological material) Results from method for determining LOD	Enter the value, or "n/a"
Evaluation of Assay Cut-off Value (diagnostic assay)	Target biological material present in the range (# to #) were used to determine the performance of the cut-off value.	Results from method for evaluating assay cut off value.	Enter the value, or "n/a"
Reference/ Normal value	Determine: Literature review or to generate in-house data. In house: (# of normal population samples) were evaluated.	Summary of results Provide references here	Enter the value, or "n/a"
Applicable genome region	(#) samples were sequenced.	Region of the genome in which sequence of acceptable quality was derived in the tested samples	Enter the value, or "n/a"

Characteristic	Number of samples tested	Actual Performance (complete calculations)	Acceptable value (if pre-defined)
Clinical validity	Determine: Literature review or generate in-house data.	Positive Predictive Value: TP / (TP + FP), and Negative Predictive Value : TN / (TN + FN) by sample type and/or population or Provide references here.	Enter the value, or “n/a”
Interfering substances	List those tested	Summary of results	Enter the value, or “n/a”
Matrix Equivalency, as applicable	Measured # of titrations of analyte (high, medium, low, and equivocal if applicable) in # replicates of negative reference matrix in parallel to # replicates of negative test matrix.		Enter the value, or “n/a”
Multiplex Assay Performance, as applicable	List common co-infections/ List co-detected analytes evaluated.	Describe the analytical sensitivity in the presence of co-infections/ co-detected analytes.	Enter the value, or “n/a”

Characteristic	Number of samples tested	**Actual Performance** (complete calculations)	Acceptable value (if pre-defined)
Posttest Probability/ Predictive Values (Clinical validity)	Literature review or in-house data. *NOTE:* If the validation has low power (e.g., n= 5 positive samples tested with 100% sensitivity, and 20 negatives with 100% specificity), this analysis should be delayed until there is adequate sample size to be informative.	Analyses performed on various pretest probabilities (prevalence) to be encountered during testing (e.g., 50%, 10%, 1%). Clinical sensitivity and specificity values derived above. Positive posttest probability = (Sensitivity x prevalence) / (sensitivity x prevalence) + ([1-specificity] x [1-prevalence]) Negative posttest probability = (1 – prevalence) x specificity / ([1-prevalence] x specificity) + ([1-sensitivity] x [prevalence]) -or- Provide references describing clinical validity of the assay	Enter the value, or "n/a"

4.2 **Interfering Substances:**

a. If interference is observed during these studies, the interferent should be tested by serial dilutions to determine the lowest concentration that provides interference. Assay limitations may be added to the validation summary.

b. Negative interferences:

Potential interfering substance to generate false negative result	Concentration	Volume of Positive Clinical Specimen Diluted, or titered "spike-in" specimen	Results
List here in multiple rows: e.g., hemoglobin, human anti-mouse antibody, rheumatoid factor	High end of clinical range	*[insert volume of positive or spike-in specimen here]*	# detected / total # replicates

c. Positive interferences:

Potential interfering substance to generate false positive result	Concentration	Results
List here in multiple rows: e.g., hemoglobin, human anti-mouse antibody, rheumatoid factor	High end of clinical range	# detected / total # replicates

6.0 Interpretation of Results:

5.0

5.1 **Evaluation of discrepant results:** Provide details describing any discrepant results when compared to the gold standard or expected value. Provide evidence to support whether the new method is likely true or not true result, as needed. If a decision is made to recategorize/exclude the sample from validation, additional evidence needs to be provided to support this decision.

5.2 **Limitations:** Provide a description of method limitations. Significant cross-reactivities, interfering substances, and other conditions known to affect test results should be included. This may include patient preparation (timing of collection, antimicrobial therapy, acute/convalescent sample timing, etc.) and sample handling (e.g., time, temperature, etc.) Utilize the following tables for guidance as applicable. Statements may include: 1) The method is not appropriate to determine (describe the restriction). 2) The method is not appropriate for use under (describe the condition(s) under which it should not be used). 3) The method is not appropriate as a stand-alone test.

a. Mitigating Processes: Describe processes used to mitigate the assay limitations (e.g., reject certain specimen types, reflexive testing for certain results, assay always performed in conjunction with other assays).

5.3 **Disclaimers** (as applicable): Provide pertinent method limitations to be included on the final test report.

5.4 **Fast Tracked test QA monitoring plan** (as applicable): For tests that have been fast tracked due to urgent public health needs, describe the metrics to be monitored and the additional evidence of test performance characteristics to be gathered. Include the timeline for gathering this evidence.

7.0 Statement of Suitability:

6.0

6.1 The method validation of (method name) has been completed according to the documented plan. The (method name) meets all the acceptance criteria and is approved for use in the (Insert name) Laboratory.

6.2 The method validation is applicable to the documents listed within the Method Validation Plan. Subsequent revisions involving technical changes to the procedure may require additional validation.

8.0 Responsibilities

Position	Responsibility
[insert title]	Is responsible for preparing the Method Validation Summary Report .
[insert title]	Is responsible for review and approval of the Method Validation Summary Report prior to testing.
[insert title]	Is responsible for performing the Method Validation Summary. Report.
[insert title]	Is responsible for review and approval of the Method Validation Summary Report upon completion.
[insert title]	Is responsible for Document Management.

9.0 Appendices

7.0

7.1 Appendix A—Summary Data

7.2 Appendix B—Specimen Stability

10.0 Revision History

Rev #	DCR #	Change Summary	Date
[insert laboratory-specific revision number here]	*[insert laboratory-specific document control number here]*	*[insert laboratory-specific change summary here]*	*[insert laboratory-specific revision date here]*

11.0 Approval

Approved By: Date:

Author

Print Name and Title

Approved By: Date:

Team Lead/Supervisor

Print Name and Title
Approved By: Date:
Quality Manager
Print Name

Appendix A—Summary Data

Please provide a line-by-line listing of all samples and results, and relevant summary data. Excel spreadsheets are acceptable.

a. Raw data: Please enter file path location of raw data here: _____.

NOTE: Please retain all validation data (worksheets, controls) and have available for review

for the life of the procedure plus two years (CLIA).

b. Isolate stability Attestation form (as applicable)

Appendix B- Specimen Stability

1. **Specimen Stability:** The table below is used to document specimen stabi study conditions and results for various conditions of specimen shipping, handling, and storage.

Specimen Condition Evaluated	Number of samples tested	Number of time points used	Temperatures selected for study	Performance (complete calculations)	Acceptability criteria from study plan
Plasma, Refrigerated, 3 days (Qualitative Assay)	(# of positive samples and # of negative samples) were measured.	Samples were measured in (#) time points as follows: (#), (#), (etc.).	Samples were stored in the following temperatures/ temperature ranges: (2°C - 8°C).	Qualitative: (%) concordance with timepoint zero (small sample size) -AND- Quantitative analysis of unreported raw values, as applicable (e.g., Ct value): acceptable mean change from baseline	*[insert acceptability criteria from specific study plan here]*
Plasma, Refrigerated, 3 days (Quantitative Assay)	*[insert number of samples tested here]*	*[insert number of time points used here as applicable]*	*[insert te mperatures selected for study here as applicable]*	Mean value within ± % of nominal value at concentrations selected	*[insert acceptability criteria from specific study plan here]*

Specimen Condition Evaluated	Number of samples tested	Number of time points used	Temperatures selected for study	Performance (complete calculations)	Acceptability criteria from study plan
Freeze and thaw stability (as applicable to mimic intended specimen handling conditions)	(# of positive samples and # of negative samples, as indicated) were measured.	Not applicable	(# of times frozen at -80°C and then thawed [i.e., # of freeze-thaw cycles])	Qualitative: __% concordance -AND- Quantitative analysis of unreported raw values (e.g., Ct value): acceptable mean change from baseline. Quantitative: Mean value within ± % of nominal value at concentrations selected.	*[insert acceptability criteria from specific study plan here]*
Respiratory sample as swab, birth defects, stool sample s	Specimens in the form of isolates	Not applicable	*[insert temperatures selected for study here as applicable]*	Not applicable	*[insert acceptability criteria from specific study plan here]*